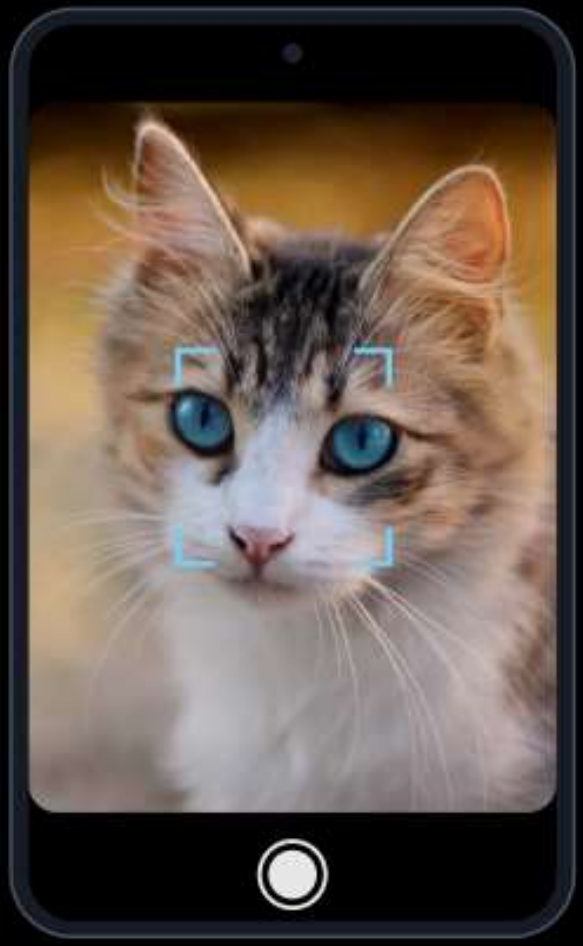

Gradient Ascent



How did it know which way to turn the lens?

Photo: AdinaVoicu (Pixabay), CC0. Focus is simulated.

This week: how machine learning learns.

Today: gradient ascent

Next lesson: maximum likelihood



By the end, you will be able to:

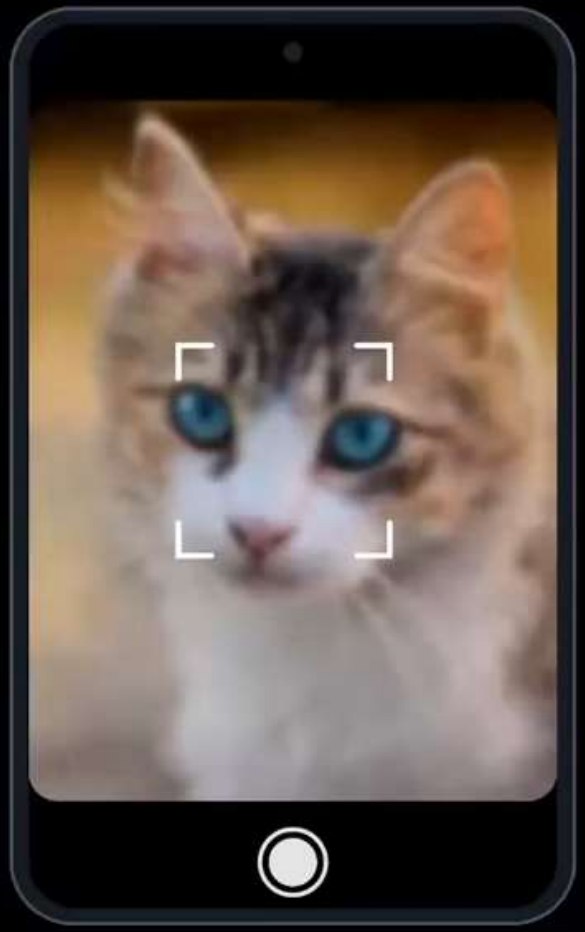
1. Understand what a gradient is.

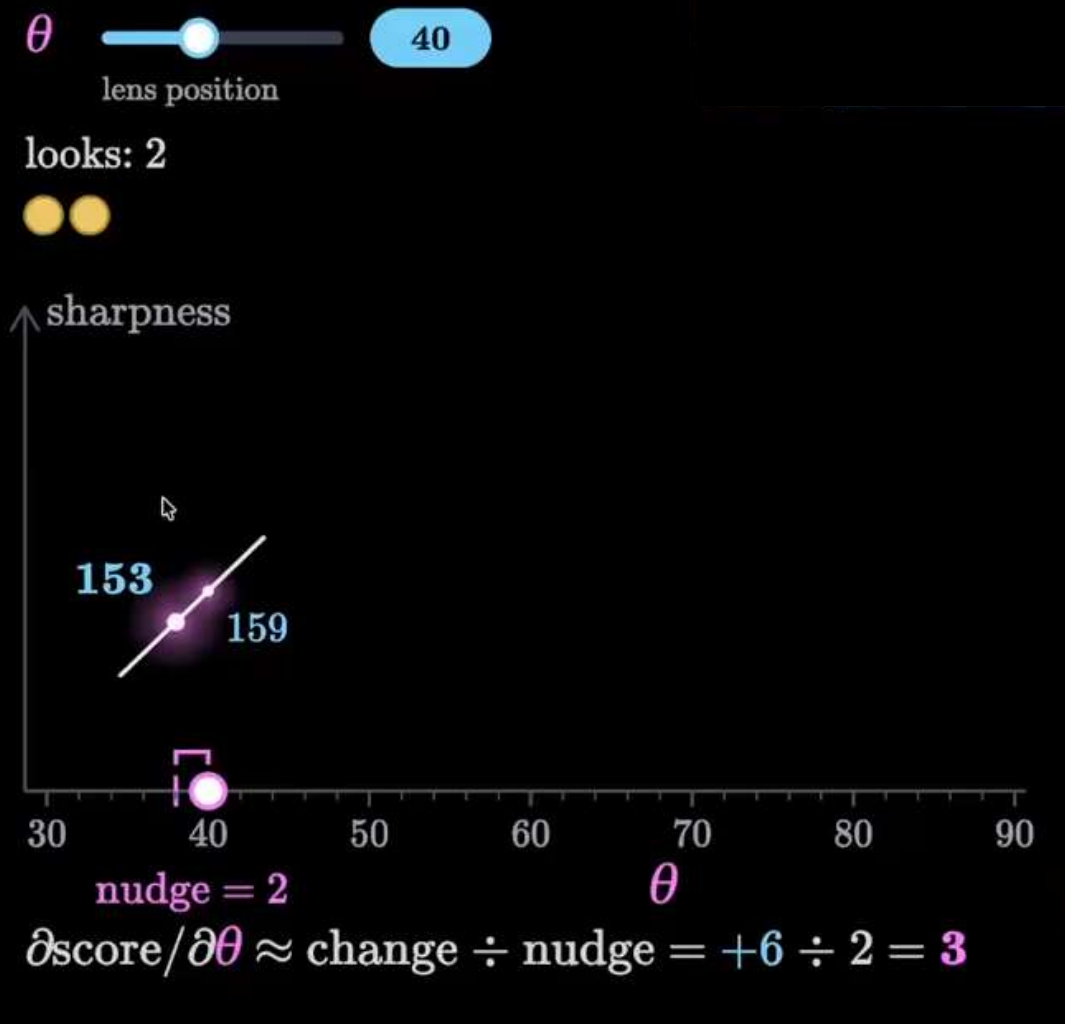
2. Use hillclimbing (aka gradient ascent) to maximize.

3. Ask a computer for a gradient.



Nudge the lens





Let's get formal!





$$\text{gradient} \approx \frac{\text{rise}}{\text{run}}$$





$$\text{gradient} \approx \frac{\text{score_after} - \text{score_before}}{\text{nudge}}$$

$$\frac{\partial\,\text{score}}{\partial\theta}$$



a measure of how well you are doing

$$\text{gradient} \approx \frac{\text{score}(\theta + \text{nudge}) - \text{score}(\theta)}{\text{nudge}}$$

a small move of θ, for a second look

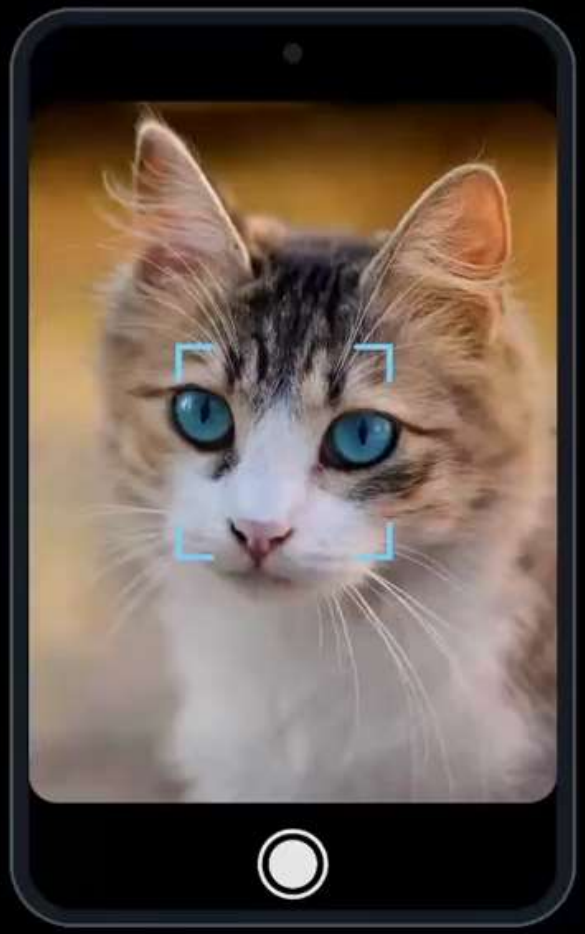
we ascend

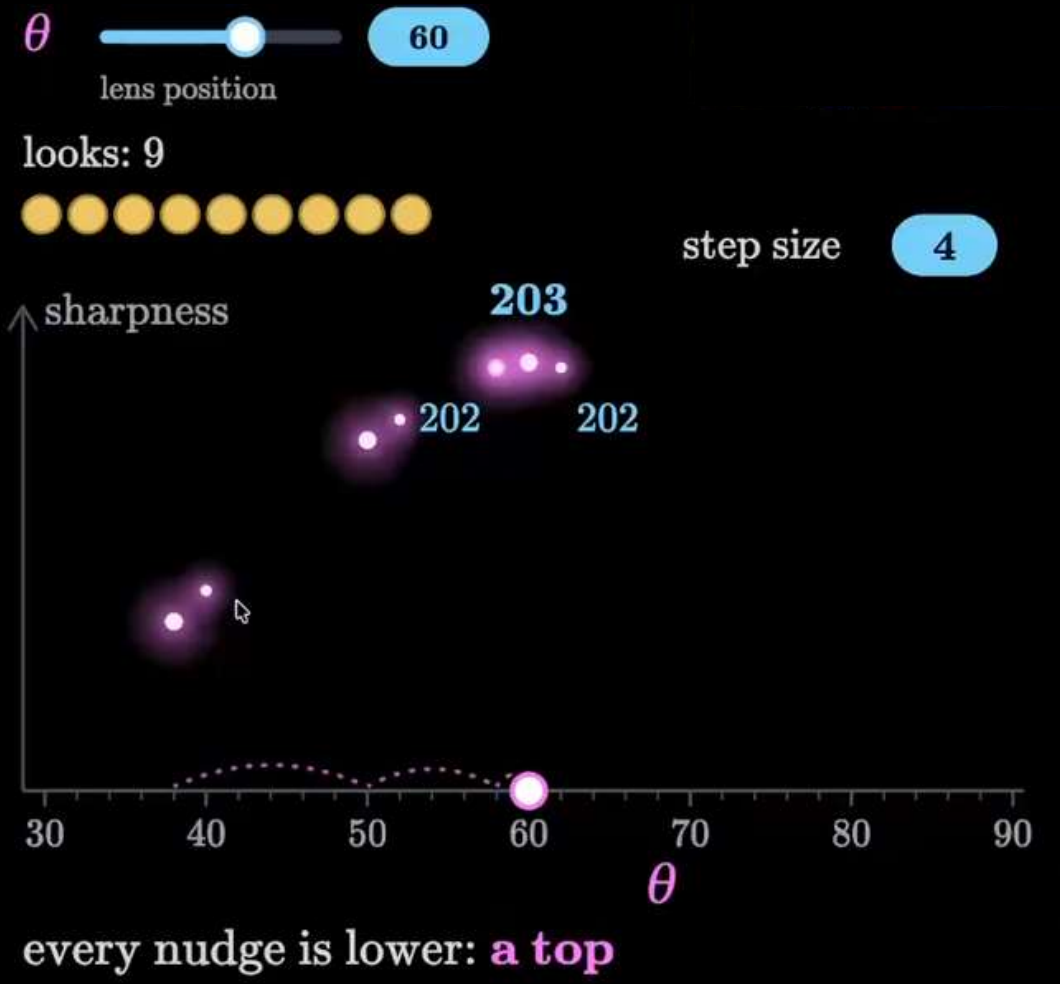


$$\text{new } \theta = \theta + \text{step_size} \cdot \frac{\partial \, \text{score}}{\partial \theta}$$



When to stop

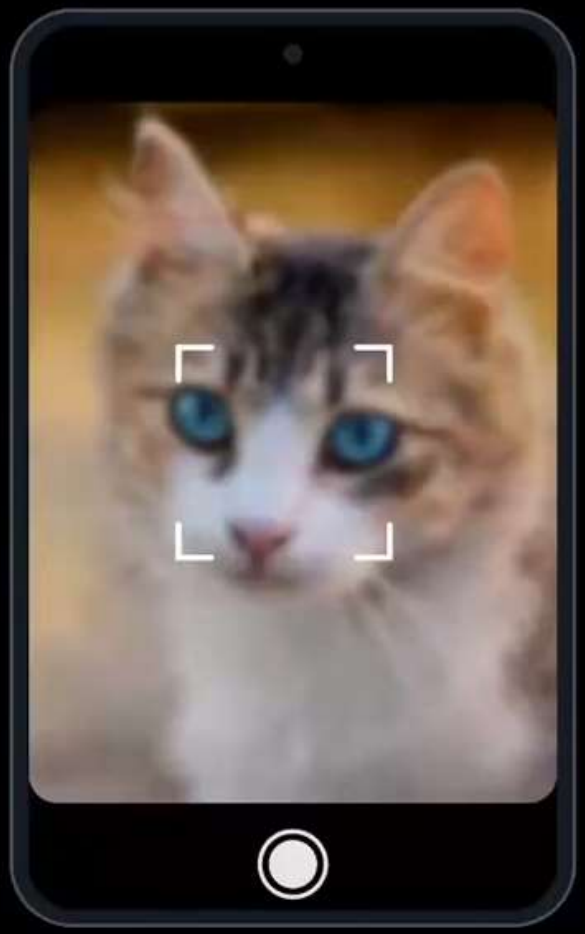


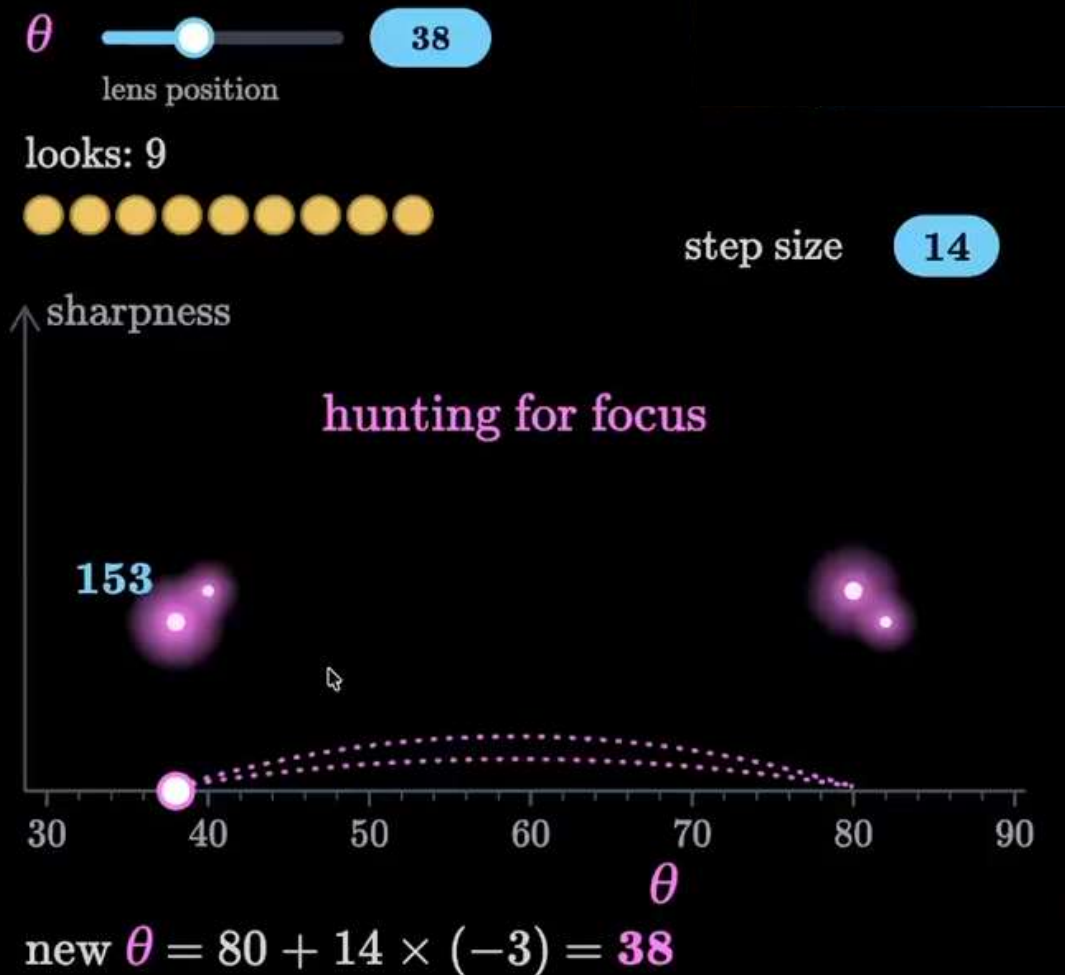

θ
60
lens position
looks: 9
step size
4
sharpness
203
202
202
30
40
50
60
70
80
90
θ
every nudge is lower: a top

What if your step size is too large?

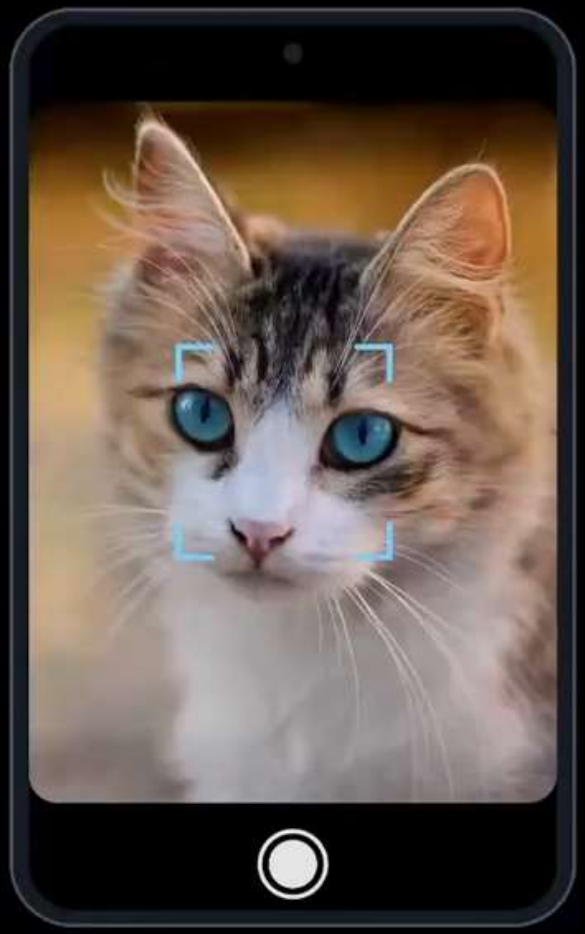


Step size 14




θ
38
lens position
looks: 9
step size
14
sharpness
hunting for focus
153
30
40
50
60
70
80
90
θ
new θ = 80 + 14 × (−3) = 38

What the camera never saw





The missing piece:

often it's easier to get the gradient than to look twice.



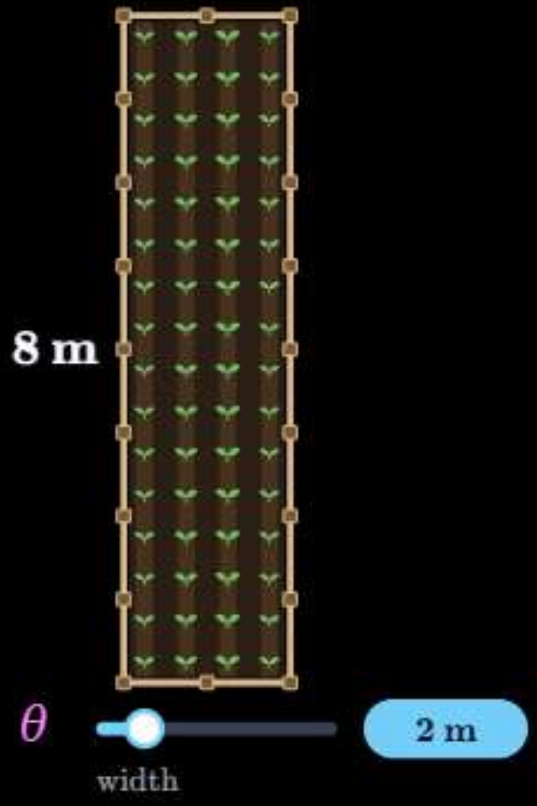
20 meters of fence

You guess: what shape is biggest?



$$2 + 8 + 2 + 8 = 20$$

The knob: θ is the width

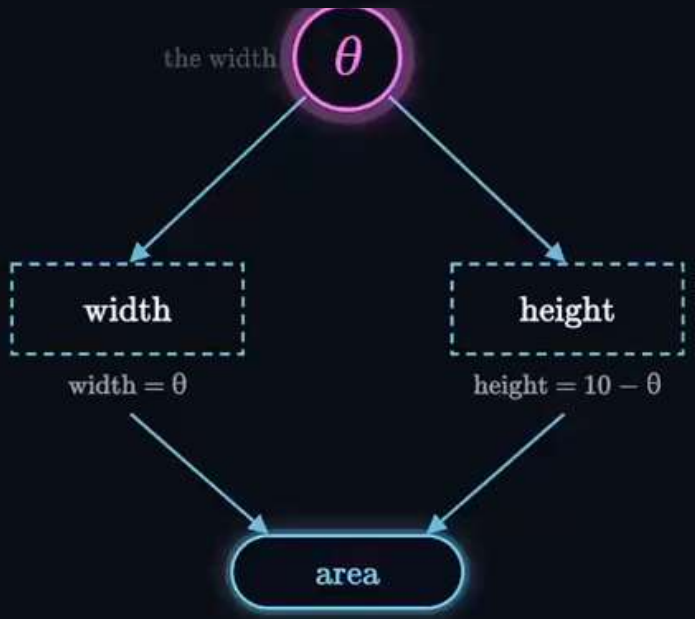


$$2\theta + 2 \times \text{height} = 20$$

$$\text{height} = 10 - \theta$$

Code Agent ✓ Done

Find the slope with respect to θ.



$$\frac{\partial \text{area}}{\partial \theta} = 10 - 2\theta$$

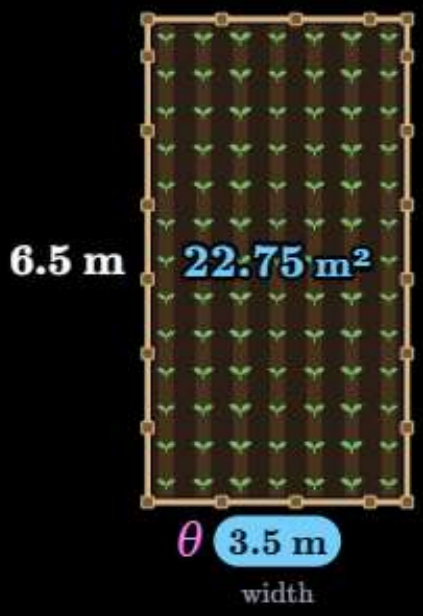
You: know you want $\partial \text{area} / \partial \theta$, and what it means.

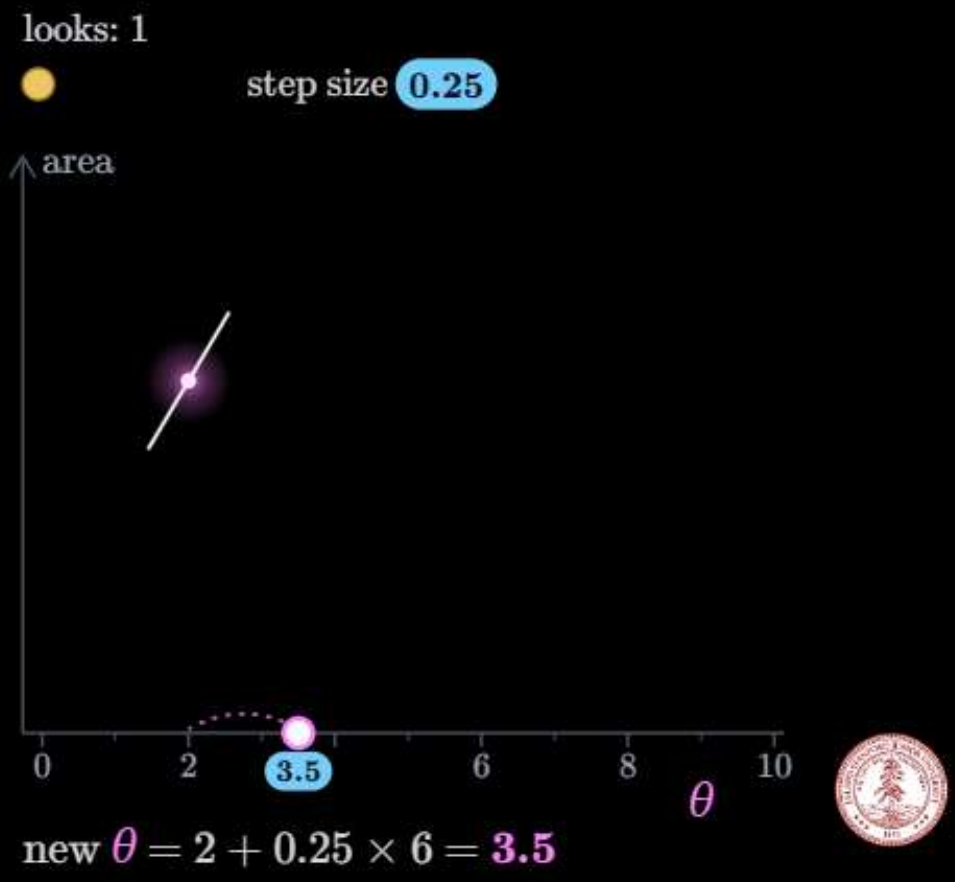
θ: width

Computers: work it out (PyTorch, 2016).

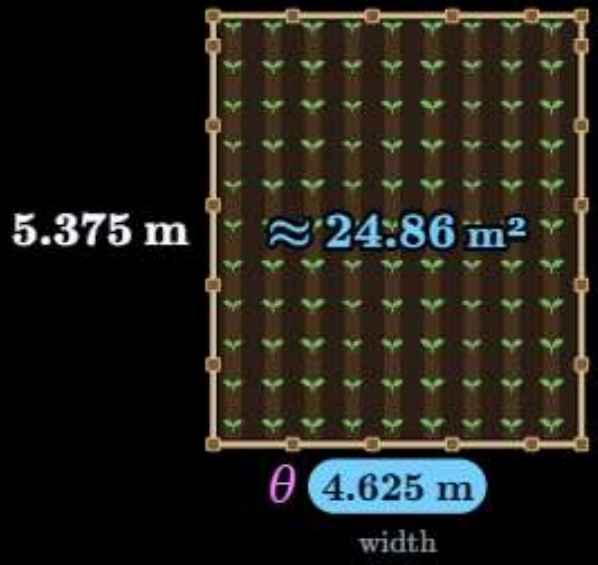


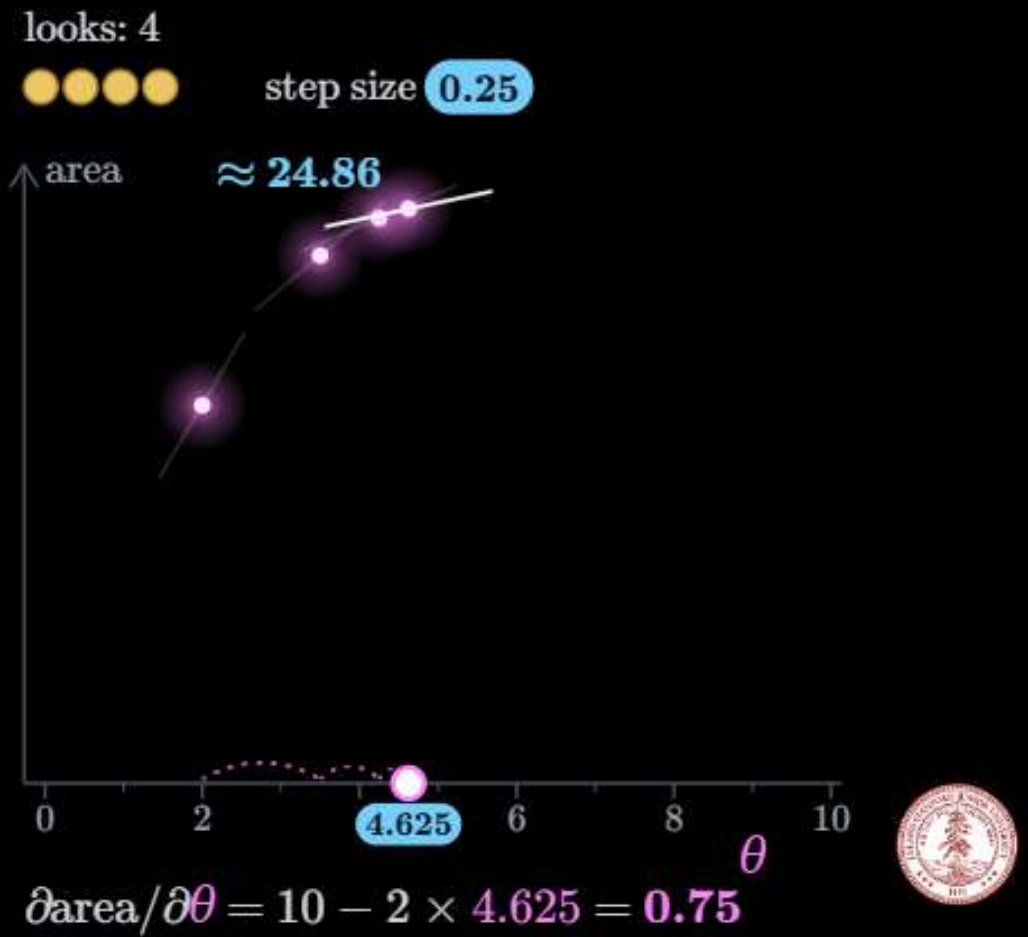
Climb with the formula





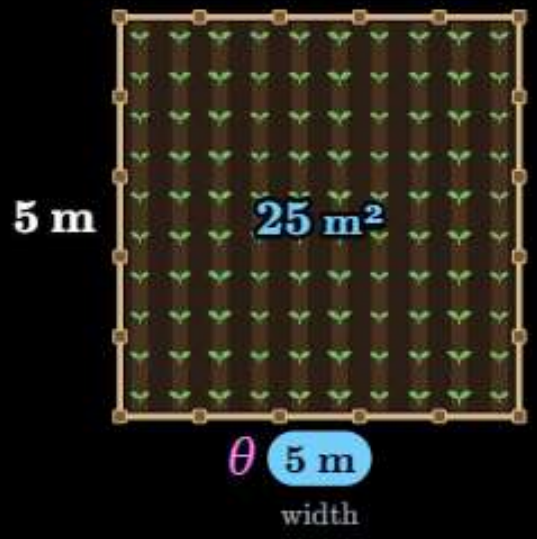
Climb with the formula





$$\partial \text{area}/\partial\theta = 10 - 2 \times 4.625 = \mathbf{0.75}$$

The whole hill





$5 \times 5 = 25$

Two knobs: lost in the fog



Photo: King of Hearts, 2013, CC BY-SA 4.0, Wikimedia Commons

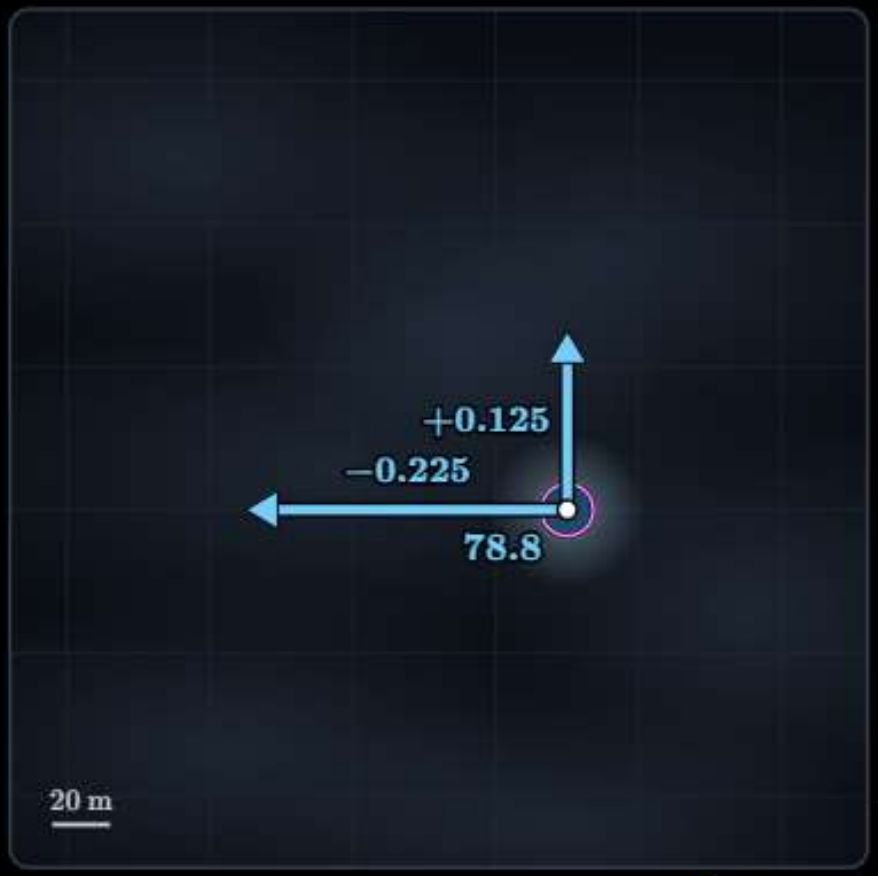
One query: height and uphill

θ_1 1825 m
east

θ_2 500 m
north

altimeter
78.8 m

looks 1



the query says
$\partial \text{height}/\partial\theta_1 = -0.225$
$\partial \text{height}/\partial\theta_2 = +0.125$

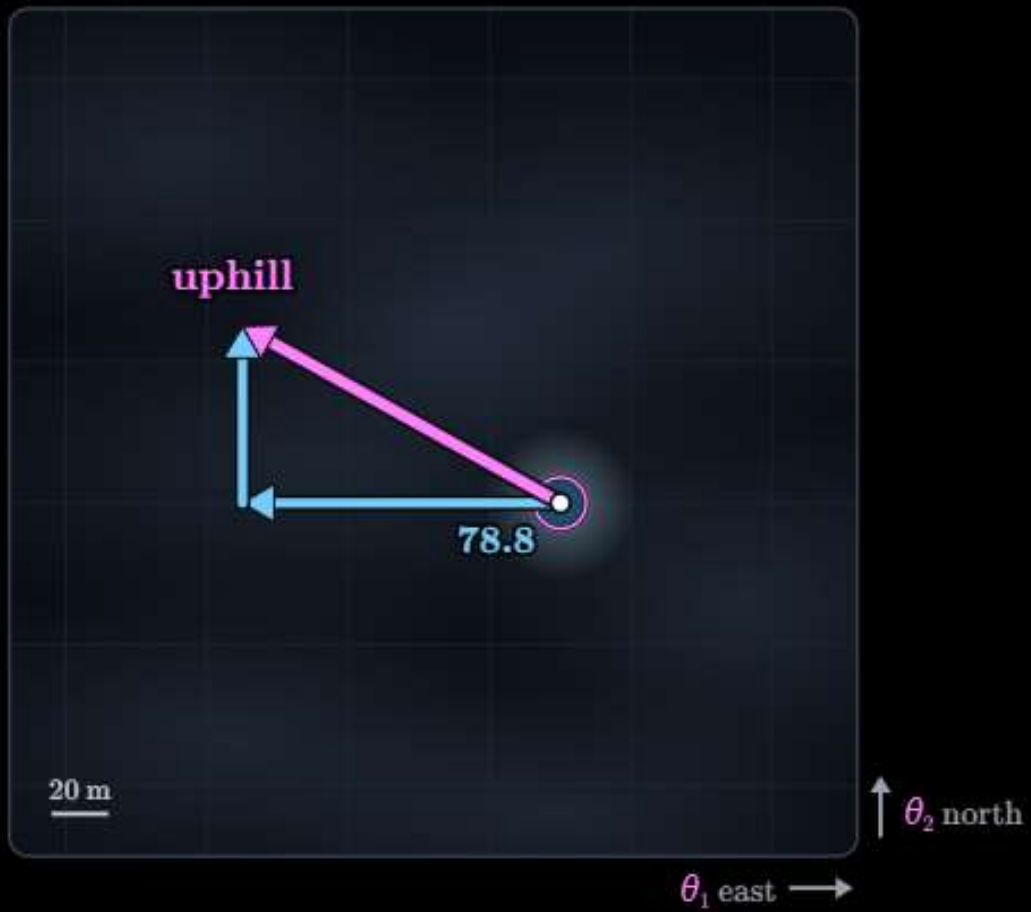
Two parts, one arrow

θ_1 1825 m
east

θ_2 500 m
north

altimeter
78.8 m

looks 1





The **gradient**: one ∂ per knob.

$$\left(\frac{\partial \text{height}}{\partial \theta_1}, \frac{\partial \text{height}}{\partial \theta_2} \right)$$

θ_1 east

θ_2 north

Together they point uphill.



Hop, query, repeat

θ_1 1712.5 m
east

θ_2 562.5 m
north

altimeter
104.7 m

looks 2



$\text{hop} = \text{step size} \times \partial\text{height}/\partial\theta$

θ_1 $500 \times -0.225 = -112.5$

θ_2 $500 \times 0.125 = 62.5$



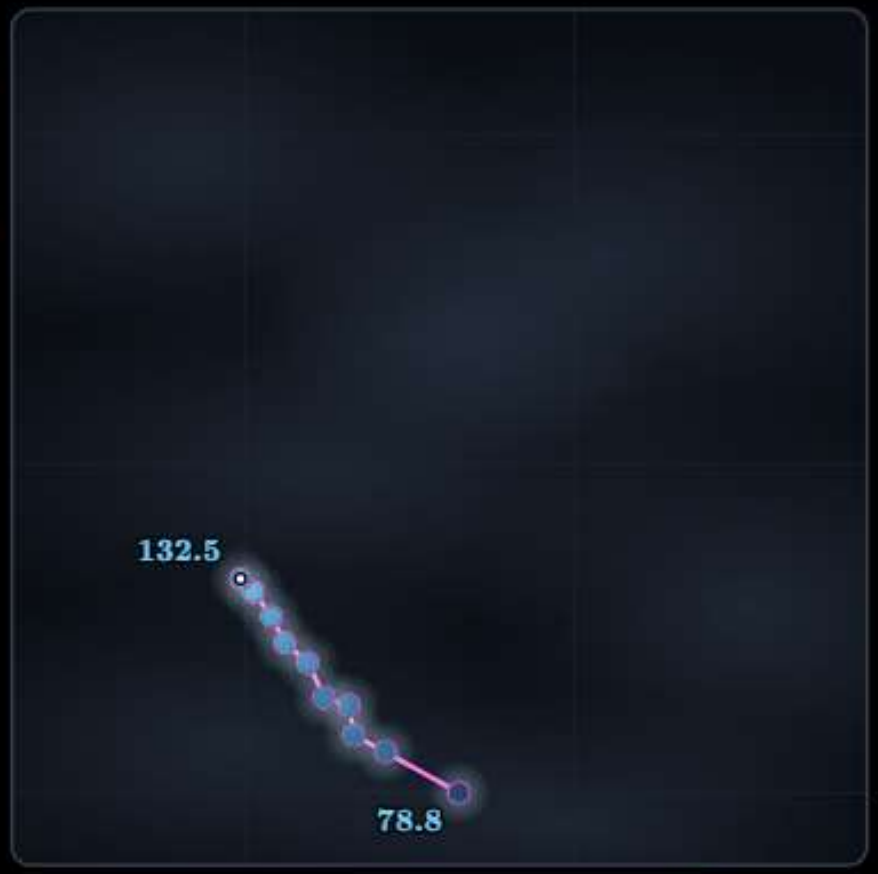
Hop, query, repeat

θ_1 1492.5 m
east

θ_2 825 m
north

altimeter
132.5 m

looks 10
per hop



The arrow goes flat

You guess: the highest?

θ_1 1492.5 m
east

θ_2 825 m
north

altimeter
132.5 m

looks 10



the slope here
$\partial \text{height}/\partial\theta_1 = 0$
$\partial \text{height}/\partial\theta_2 = 0$

flat: a top

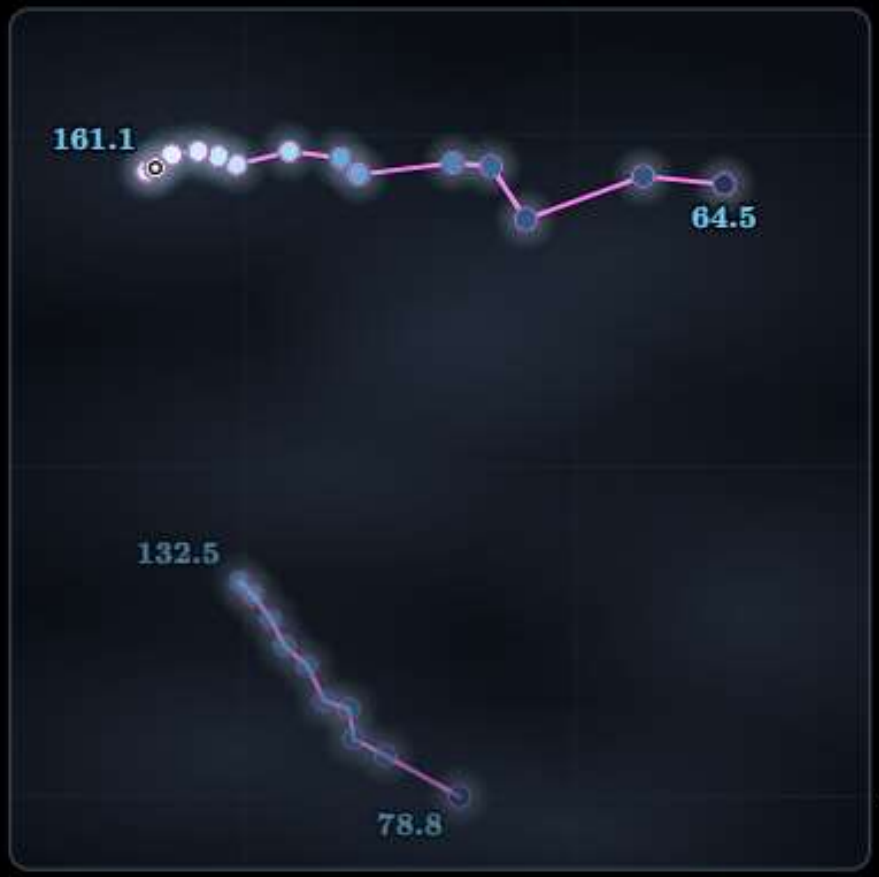
A different start

θ_1 1365 m
east

θ_2 1450 m
north

altimeter
161.1 m

looks 24
per hop



161.1 > 132.5
higher

What the climber never saw

looks 24



Many tops.

Two almost tie.

- $\dfrac{\partial\,\text{score}}{\partial\theta} \approx \dfrac{\text{score}(\theta + \text{nudge}) - \text{score}(\theta)}{\text{nudge}}$
- $\text{new}\ \theta = \theta + \text{step size} \times \dfrac{\partial\,\text{score}}{\partial\theta}$
- Step size: not too small, not too big
- Stop: every nudge goes down
- Ask for: $\partial\,\text{score}/\partial\,\theta$



Next lesson

The score: the chance of your data.

A formula: ask for its gradient.



Footnote



In practice, AI flips it: gradient descent on a loss. Same climb.

That's all folks!



Now you know what your phone is doing.







Photo: AdinaVoicu (Pixabay), CC0. Focus is simulated.